

Rapid Neural and Behavioral Adaptation in a Updated Theory of the Cognitive Map and its Neural Implementation

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ABSTRACT

Animals rapidly and flexibly adapt their behavior by forming an internal cognitive map. The cognitive map has been studied at multiple levels, including normative, representational, and mechanistic accounts. Here, we propose a unified theory linking these levels, filling in missing pieces left unanswered by studies at any single level. In our model, CA3 uses temporally symmetric behavioral time scale plasticity (BTSP) to encode spatiotemporal proximity experienced during exploration, CA1 uses temporally asymmetric BTSP to encode a perceived-salience-weighted successor representation, and downstream areas learn successor feature predictions that can guide behavior. Through simulations of a rodent maze-learning task, we show that the network rapidly learns a predictive map, supporting offline replay which further strengthens the learned predictive representations. Behavioral simulations show how these processes give rise to rapid behavioral adaptation, including motivational-state-dependent choice and one-trial avoidance learning.

Introduction

Animals adapt quickly to their experience in spatial environments. This ability is closely tied to the concept of a "cognitive map"—a putative representation that receives locally available information, integrates it with internal state information, and generates signals that guide behavior¹⁻³. Classic maze experiments show that rodents rapidly form the cognitive map to make efficient decisions (Fig. 1a), motivating three central questions: **(1) What is learned?** In what form is the cognitive map represented in the brain? **(2) How is it learned?** What neural mechanisms give rise to the cognitive map? **(3) How is it used?** How does the output of the cognitive map contribute to evaluating different possible choices and guiding behavior?

To address the question *What is learned?*, we build on the account of the cognitive map provided by the successor representation (SR)^{4,5}. Rather than encoding environments purely in terms of spatial distance, the SR encodes the expected future occupancy of each location based on the animal's movement history (Fig. 1b, top). Although SR is supported by several experimentally observed properties of hippocampal place cells⁵, including the predictive skewing of place fields^{6,7}, important aspects remain unaccounted for⁸. In particular, SR does not explain the stronger hippocampal encoding of behaviorally significant but infrequently visited locations, such as those associated with aversiveness⁹, motivational salience¹⁰, or novelty¹¹.

These factors, collectively termed perceived salience (PS)¹², provide a crucial input for the hippocampus's rapid and flexible learning. Experimental evidence shows that hippocam-

pal representations are modulated by PS, including increases in the number of place cells encoding salient locations¹³⁻¹⁶ and accelerated formation of those fields¹⁷. Together, these findings indicate that the hippocampus integrates PS signals with spatial inputs, producing a PS-weighted SR that is more adaptive than a purely spatial SR.

To answer *How is it learned?*, we focus on behavioral time-scale synaptic plasticity (BTSP)¹⁸ as a biologically plausible local mechanism that can acquire a PS-weighted SR. BTSP is notable for producing large synaptic changes linking neural activity across a time-scale of seconds, allowing neurons activated within a behavioral timescale to be linked. These changes are strongly modulated by novelty, salience, and reward¹⁹⁻²², making BTSP a compelling candidate for learning PS-weighted SR.

A distinct form of learning occurs when the animal is not actively exploring but is instead at rest or asleep. During this "offline" phase, place cells are spontaneously reactivated in sequences that recapitulate trajectories through the environment²³. This phenomenon, known as hippocampal replay, is thought to contribute to the stabilization and consolidation of learning acquired during physical exploration of the environment²⁴⁻²⁶. Remarkably, these sequences can represent trajectories that the animal has never physically traversed²⁷, suggesting that replay helps complete the internal spatial map^{28,29}. However, the precise mechanisms by which replay influences synaptic change are not yet fully understood. In our work, we propose how replay can support consolidation through spike-timing-dependent plasticity (STDP).

Finally, to answer *How is it used?*, we consider how the

learned PS-weighted SR guides decisions. The original SR is limited in that it does not explain how feature-specific factors, rather than entire states, influence value⁸. For example, water becomes valuable only when the animal is thirsty, selectively increasing the value of states containing water. This requires a mechanism that converts feature-dependent signals into state values. To address this, we incorporate successor features (SFs)³⁰ previously proposed in a machine learning context. Unlike the SR, which predicts the *future occupancy of states*, SFs predict the *future occurrence of features* (Fig. 1b, bottom). We propose that the PS-weighted SR is translated into SFs in downstream regions of the hippocampus. These predicted features are then combined with feature-specific value signals that dependent on the animal's transient needs, to eventually compute the overall value of each state and guide behavior.

Combining these components, we propose a unified framework of the cognitive map (Fig. 1c) and map it onto its neural implementation and behavioral consequences. In our model, the hippocampus integrates spatial input with perceived-salience signals to form a PS-weighted successor representation, which is then conveyed to downstream areas that compute successor features and combine them with motivational signals to estimate state values (Fig. 1d). To demonstrate how this representation emerges and supports rapid behavioral change, our simulations employ three distinct phases: an online exploration phase, during which BTSP updates the hippocampal circuitry (Fig. 1e), and an offline replay phase, during which spontaneous replays strengthen and propagate predictive structure (Fig. 1f), and a behavioral phase in which the learned predictive map supports flexible value computation and guides behavior (Fig. 1g).

Results

Formal Statement of Theory, Its Neural Implementation, and Simulation Approach

Following⁴, the successor representation (SR) for a starting state s and a target state s' is the expected discounted future occupancy of s' across time t , given $s_0 = s$:

$$M(s, s') = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \mathbb{I}(s_t = s') \middle| s_0 = s \right], \quad (1)$$

where $\gamma \in [0, 1]$ is a discount factor. Thus $M(s, s')$ weights nearer visits more heavily than distant ones.

The SR then allows us to characterize successor features (SFs), which represent predictions of future occurrence of environmental features (e.g., cues, rewards, or hazards)³⁰. Let $\phi(s', f)$ denote the expected presence of feature f at state s' . The SF for feature f at start state s is defined as the expected discounted future occurrence of that feature:

$$\begin{aligned} \psi(s, f) &= \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t, f) \middle| s_0 = s \right] \\ &= \sum_{s'} M(s, s') \phi(s', f). \end{aligned} \quad (2)$$

We further incorporate perceptual salience (PS), $\omega(s')$ into our framework. First, we define the PS-weighted SR,

$$M^{\text{PS}}(s, s') = \omega(s') M(s, s'). \quad (3)$$

That is, PS amplifies states by $\omega(s')$ in the SR. With this scaling, the SF can be expressed as a linear map from M^{PS} :

$$\psi(s, f) = \sum_{s'} M^{\text{PS}}(s, s') \cdot W_{\psi, M^{\text{PS}}}(s', f), \quad \text{where} \quad (4)$$

$$W_{\psi, M^{\text{PS}}}(s', f) = \frac{\phi(s', f)}{\omega(s')}. \quad (5)$$

Finally, the value of state s is defined as the sum of successor features weighted by their value:

$$V(s) = \sum_f \psi(s, f) v(f), \quad (6)$$

where $v(f)$ is the value of feature f at the time when $V(s)$ is estimated. This allows the same predictive representation ψ to flexibly support different decisions, depending on the agent's internal motivational state.

We link these theoretical quantities to neural processes and behavior as follows:

- An estimate of the **PS-weighted SR**, $M^{\text{PS}}(s, s')$, is acquired through BTSP in hippocampal areas CA3 and CA1 during exploration and is further reinforced through replay-driven plasticity.
- **Perceived salience**, $\omega(s)$, is determined based on the intensity, novelty, and motivational significance of features present at state s as represented in the brain by motivational signals projecting to area CA1, and is incorporated by modulating BTSP.
- The **mapping from PS-weighted SR to SF**, $W_{\psi, M^{\text{PS}}}$, is instantiated by learned synaptic weights from the CA1 layer to downstream feature-encoding populations.
- **State value**, $V(s)$, is computed in extra-hippocampal circuits integrating feature predictions with feature-specific value signals $v(f)$, which depend on the animal's internal needs during behavior.
- The resulting $V(s)$ guides **decision making**: when selecting actions, animals preferentially transition to states with higher $V(s)$.

In the following sections, we computationally simulated how our framework emerges with three distinct paradigms of simulation: the online, offline and behavior phases (Fig. 1e–g; see *Methods* for details).

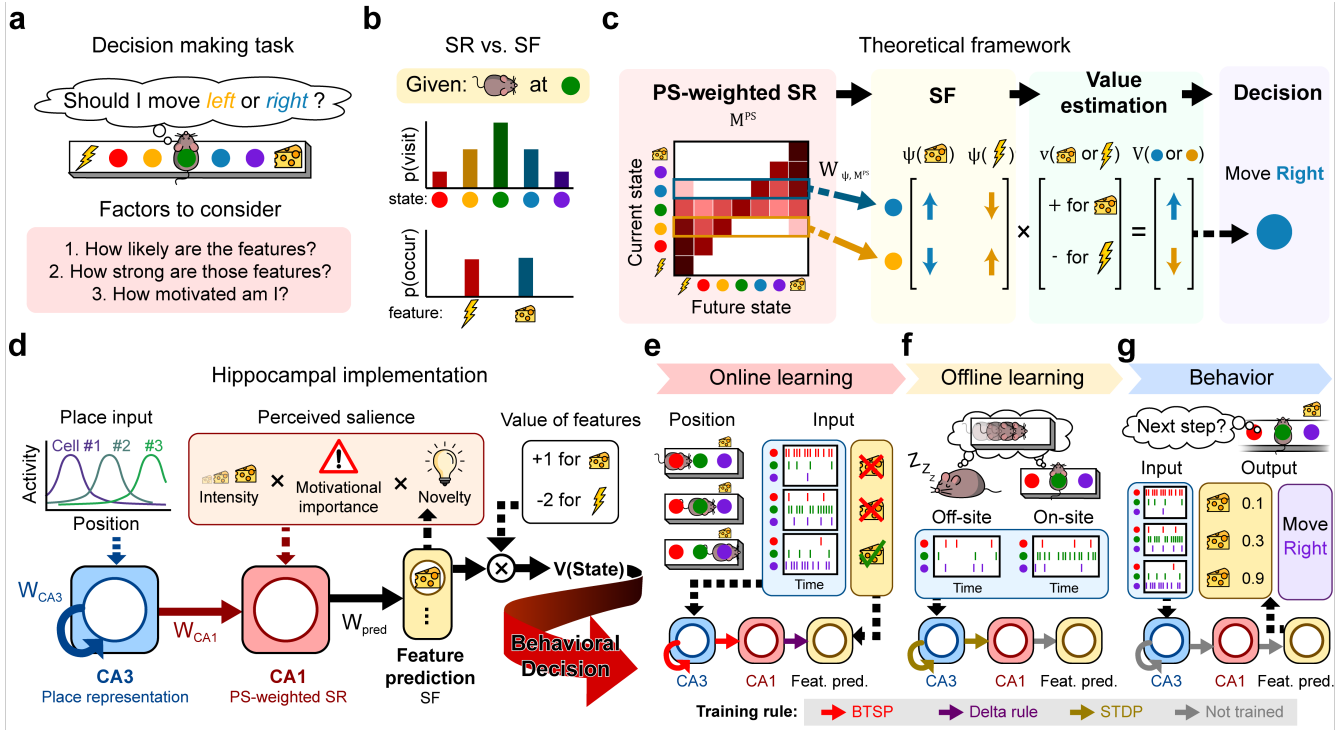


Figure 1. **a**, Rodent decision-making in a maze. **b**, Two ways of representing a state: successor representation (SR) and successor features (SF). **c**, Schematic of the theoretical framework. A PS-weighted SR is relayed to predict successor features, which are then used to compute value and guide decisions. **d**, Hippocampal implementation of the model. CA3 receives place input and encodes place representations. CA1 receives input from CA3 along with perceived salience, and encodes the PS-weighted SR. This is passed to the feature prediction layer, which encodes SF and is ultimately used to determine the subjective value of states. **e–g**, Three distinct phases of our simulations. **e**, *Online learning*: the animal actively explores the environment and encounters features. Place input varies with position, and hippocampal weights are updated using the BTSP rule, while the feature-prediction weights are trained using an error-correcting delta rule. **f**, *Offline learning*: simulating rest or sleep. Place input is held constant, and hippocampal weights are updated using symmetric STDP while the feature-prediction weights are fixed. **g**, *Behavior simulation*: the animal performs value-based decision making in the learned track. No weights are updated. Instead, the animal exploits the learned network to calculate the value of possible choices and make sequential decisions.

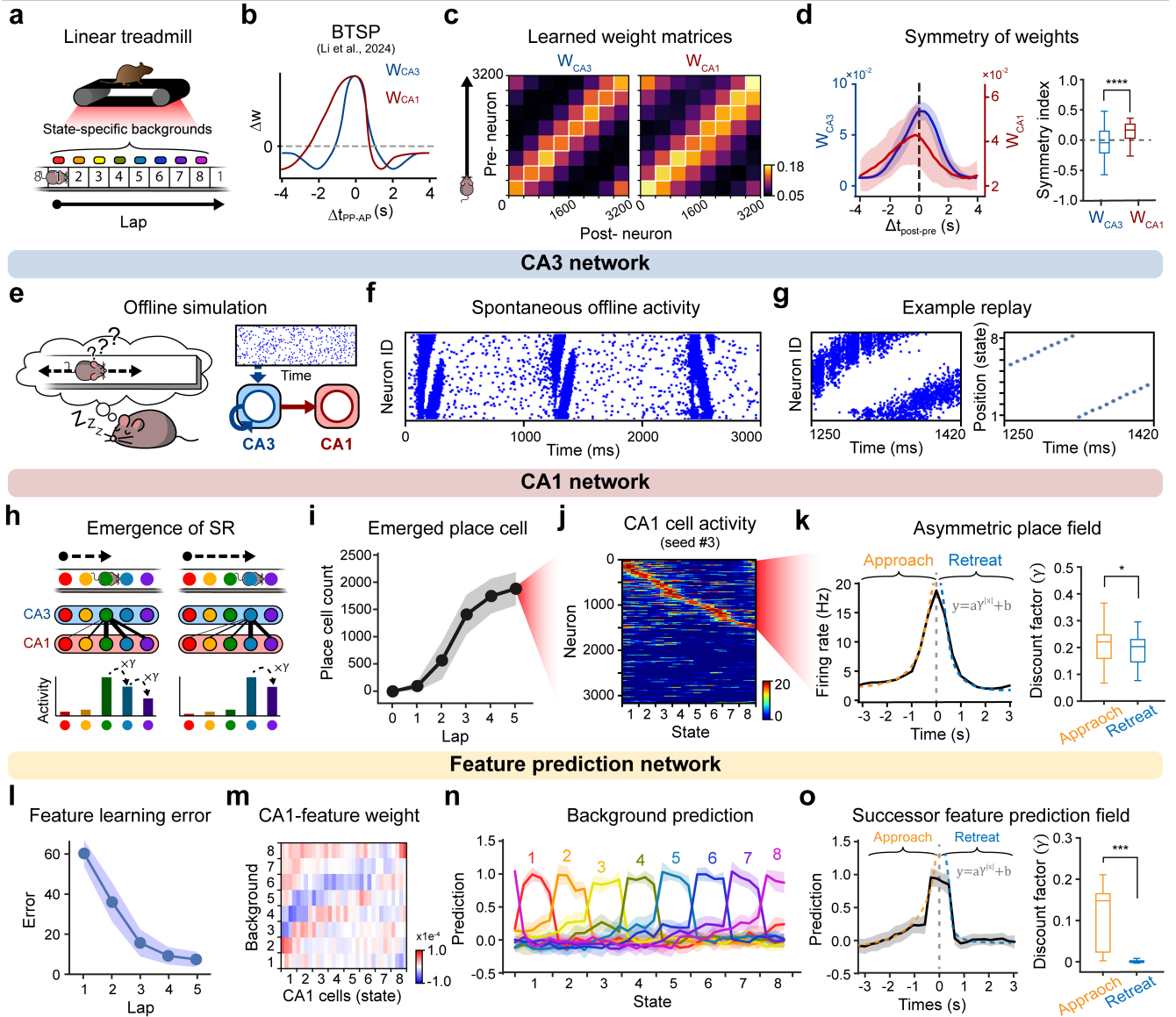
Specialized representations of CA3 and CA1 regions emerge through BTSP

First, we demonstrated that the hippocampus can implement our normative framework in an environment without salient outcomes, with BTSP as the synaptic learning mechanism. As a simplified brain circuit model, we constructed a three-component network: a CA3-like recurrent layer with weights W_{CA3} , a CA1-like feedforward layer with weights W_{CA1} , and a downstream “feature-prediction” layer, all initialized randomly. Aside from W_{CA3} , each CA3 neuron received place-specific inputs based on its assigned fixed place field centered at a specific location that evenly tiles the environment. On the other hand, CA1 neurons show non-spatial specific activity upon initialization. We modeled navigation as uniform movement through the assigned CA3 place field centers by dividing the linear treadmill into eight consecutive states (states 1–8). Each state is associated with a neutral “background” feature used solely to distinguish spatial locations (Fig. 2a). During this online navigation, W_{CA3} was trained using the BTSP learning rule with a symmetric time kernel, whereas W_{CA1} was trained using an asymmetric kernel (Fig. 2b)³¹.

This distinction in learning kernels resulted in different structures in W_{CA3} and W_{CA1} . To illustrate this, we assigned each CA3 and CA1 neuron an ID based on the spatial position

where it exhibited peak firing. Then, we plotted W_{CA3} and W_{CA1} , sorting both presynaptic and postsynaptic neurons by their IDs (Fig. 2c). The results showed that W_{CA3} encoded spatial relationships between place fields in an allocentric-, movement history-independent manner, resulting in a symmetric structure (Fig. 2d, blue). In contrast, W_{CA1} displayed a skewed distribution: connections from place cells representing positions behind the animal’s current location were more potentiated than those ahead (Fig. 2d, red).

Next, we examined the functional consequences that ensued from these distinct representations. We first showed that online learning through the symmetric BTSP rule enabled the CA3 network to spontaneously generate replay-like activity during the following offline periods. To test this, we simulated the offline phase by delivering a uniform, non-spatial input for three seconds to the trained W_{CA3} to mimic sleep conditions (Fig. 2e). We simulated the resulting spontaneous activity using biophysically plausible parameters and an adaptive exponential integrate-and-fire neuron model³² (see Supplementary Section ??). As a result, CA3 neurons fired in sequence according to their place-field locations (Fig. 2f). These sequential events were best decoded as a linear travel along the track and therefore qualify as replay events³³ (Fig. 2g). Strikingly, the replay activity emerged after only a single lap of



exploration³⁴. In contrast, networks trained with STDP rule required approximately 400 laps before exhibiting comparable replay dynamics (Supplementary Fig. ??).

Next, we showed that the asymmetric structure of W_{CA1} endows CA1 neurons with SR-like activity patterns (Fig. 2h). To verify this, we examined the activity of individual CA1 neurons, which were randomly initialized. We classified a CA1 neuron as a place cell if its peak firing rate exceeded 15 Hz (75% of the model's theoretical maximum of 20 Hz) at any position along the track. As a result, we found that after five laps of exploration, an average of 1,867 CA1 place cells had emerged (Fig. 2i). The resulting place fields were distributed across the environment (Fig. 2j; also see Supplementary Fig. ??). As expected, these place fields were asymmetric: neurons began firing earlier as the animal approached the field's peak location and shut off more rapidly as it moved away (Fig. 2k, left). By fitting the place cell activity during the "approach" phase to the curve $y = a \times \gamma^{|x|} + b$, we estimated the empirical discount factor γ (Fig. 2k, right). The discount factor fell between 0 and 1, indicating that the network prioritized memory of the immediate to more distant past. For comparison, we calculated the discount factor during the "retreat" phase, which was lower than that of the approach phase.

Finally, we examined whether the feature prediction network successfully learned the SFs. During online simulations, W_{pred} was trained using the error correcting delta rule³⁵ to predict the presence of the eight background features associated with the track states. As expected, the mean prediction error decreased across the five laps. This indicated that as CA1 developed place fields, the prediction layer increasingly relied on them to better infer the corresponding features (Fig. 2l). By the end of the fifth lap, W_{pred} had become state-specific, with CA1 place cells preferentially projecting to the feature unit associated with their place field (Fig. 2m). Consequently, each background feature unit exhibited strongest prediction when the animal occupied its corresponding location (Fig. 2n). These learned feature fields showed anticipatory behavior as CA1 place fields (Fig. 2o).

Salient features are adaptively encoded and updated in CA1 and feature prediction network

Next, we demonstrated that the SR encoded within the CA1 layer reflects the introduction and relocation of salient outcomes. To test this, we included a salient positive feature (a food reward), in the simulation. After five familiarization laps without salient features, the reward was placed at state 4 for five laps (laps 6-10). Then, the reward was then relocated to state 8 for five additional laps (laps 11-15) (Fig. 3a).

We treated the introduction of reward as affecting two key factors (Fig. 3b). First, the motivational importance of food was set to 3, compared with 1 for background states. This significantly increased the perceived salience of the reward location (Fig. 3c). Second, the animal's travel speed was halved when passing through the reward state^{18,22}.

We next examined whether the SR encoded within W_{CA1} captured the salience information introduced by the reward. Unlike W_{CA3} that was not meaningfully altered (Fig. 3d), W_{CA1} adapted rapidly to changes in reward location: after lap 6, weights projecting from CA3 neurons representing state 4 were selectively strengthened, and after lap 11, weights from neurons representing the new reward location at state 8 increased sharply (Fig. 3e). This incorporated diverse forms of place-field adaptation, reflecting the versatility of BTSP depending on perceived salience (Supplementary Fig. ??). For example, after reward relocation, some CA1 neurons acquired new fields at the reward site, whereas others shifted their existing fields toward the new location (Fig. 3f). Collectively, the CA1 population reorganized to over-represent the vicinity of the reward (Fig. 3g)^{13,14,21}.

We next examined how these representational changes improved reward prediction, distinguishing between the novel (immediately following introduction or relocation) and the familiar reward exposure condition (after repeated experience) (Fig. 3h). Over-representation of the reward site increased CA1 population activity at that state upon repeated exposure (Fig. 3i). This in turn accelerated learning of the reward-state association: Prediction error for the reward decreased far more rapidly than for background features (Fig. 3j). Eventually, the network formed reliable reward predictions after a single encounter (Fig. 3k). However, our simulations showed that anticipatory prediction emerged gradually: only after being familiar with the reward, the animal showed increasing predictive activity as it approached the reward (Fig. 3l).

The network recapitulates behavioral preferences based on motivational state

We next tested how the learned framework supports decision making based on an animal's transient motivational state. We trained the networks on a two-dimensional T-maze with two distinct arms. In every odd lap, the animal started at the stem (state 1) and moved toward the left arm (state 4), whereas in every even lap it moved toward the right arm (state 10) (Fig. 4a). After six laps (three exposures to each arm), two distinct reward features were introduced at the arm ends (food and water) and the animal was exposed to each reward twice (laps 7-10) (Fig. 4b).

After lap 10, W_{CA3} encoded the structural layout of the T-maze, while W_{CA1} over-represented both arms, reflecting the presence of food and water (Fig. 4c). Correspondingly, the feature prediction layer strongly predicted each reward at its respective location (Fig. 4d).

We then simulated behavioral phase under different motivational needs for food and water (Fig. 4e). In each simulation, the animal started at the stem (state 1) and advanced one state per second. At each step, it evaluated all possible next states including neighboring states and the current state, and translated these values into movement probabilities (Fig. 4f, also see Methods). Each behavioral trial lasted up to 60 seconds; if neither goal was reached, the trial ended.

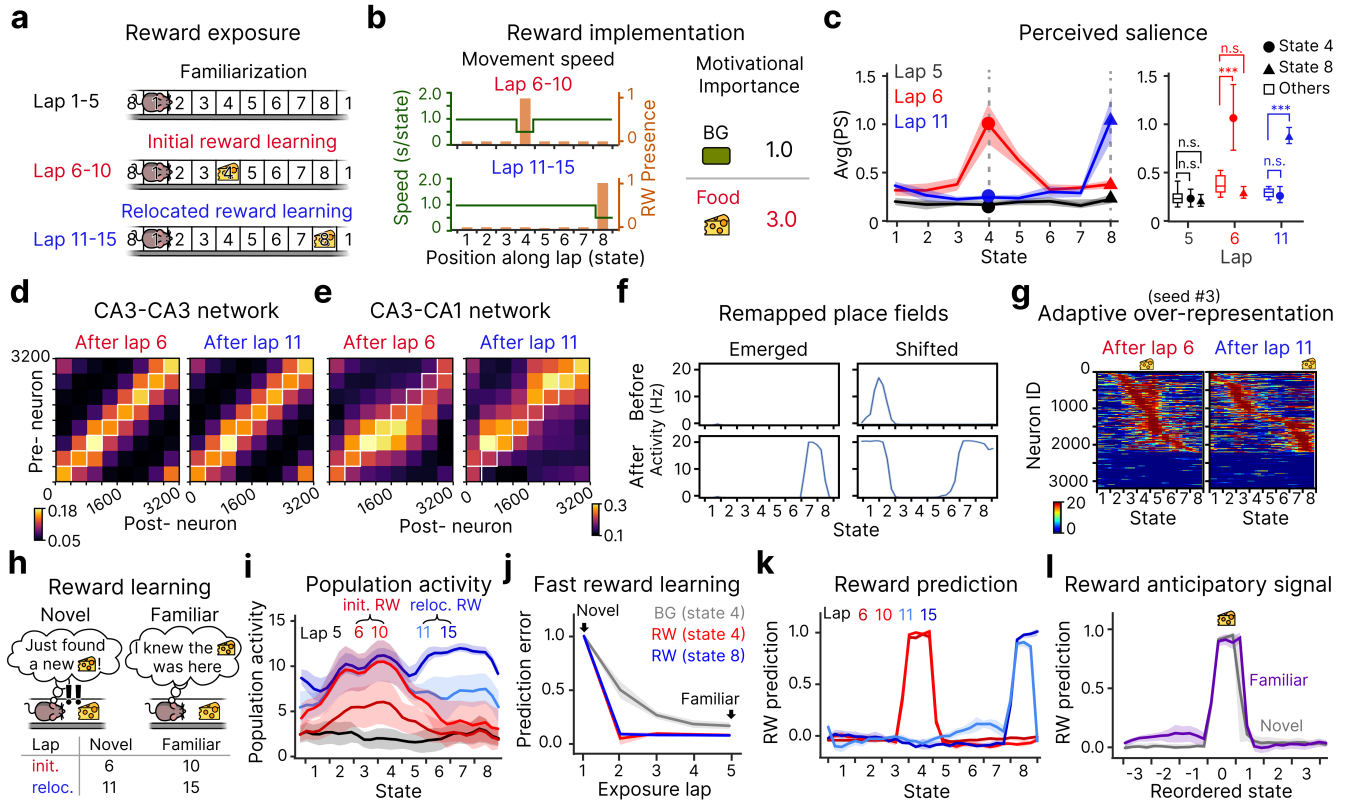


Figure 3. **a**, Simulation scheme for introducing a salient reward. **b**, Two factors modulated by the introduction of reward: decreased movement speed and the motivational importance of the reward. **c**, Left: Average perceived salience of each state on laps 5, 6, and 11. The rewarded state shows higher perceived salience when the reward is present. Data are presented as mean $\pm$ s.d. Right: Perceived salience at state 4, 5 and the rest experienced during lap 5, 6 and 11. Paired t-test, $N_{seed} = 10$, n.s. $P > 0.05$, *** $P < 0.005$. **d**, Weights of W_{CA3} after lap 6 (initial reward introduction) and lap 8 (reward relocation). Average value across 10 different seeds are plotted. **e**, Weights of W_{CA1} after lap 6 and 11. Average value across 10 different seeds are plotted. **f**, Example place field revisions after reward removal: emergence of new place cells (left), and shifts of field peaks (right). **g**, CA1 cell activity after lap 6 and 11. Place cells clustered around the reward states, which representation adapts based on reward introduction and relocation. Average value across 10 different seeds are plotted. **h**, Two different cognitive states associated with reward perception. When the reward is presented for the first time at a given location, the animal experiences it as novel (left). After repeated exposures at the same location, the reward becomes familiar (right). **i**, Population activity of the CA1 layer before reward across laps, computed as the mean firing rate averaged across all CA1 neurons. **j**, Prediction error for background features and the reward feature across laps. The animal required more laps to learn the background feature than the reward feature. **k**, Predicted presence of the food across track positions. **l**, Reward prediction aligned to the animal's movement direction, with 0 indicating the reward location. After familiarization, the animal displayed anticipatory prediction when approaching the reward. All data are presented as mean $\pm$ s.d.

We simulated three motivational conditions: (1) hungry (food need = 10, water need = 1), (2) thirsty (food need = 1, water need = 10), and (3) neutral (food need = 1, water need = 1). For each condition, we computed the Markov transition matrix (Fig. 4g). The three matrices notably differed in the state transition probabilities following state 7, the choice point in the maze. When hungry, the animal was more likely to move toward the food arm (left), whereas when thirsty it preferred the water arm (right). Under neutral conditions, movements left, right, or back to the stem were roughly balanced. Accordingly, the animal predominantly reached the food when hungry, the water when thirsty, and distributed its choices more evenly under neutral conditions (Fig. 4h-i). Finally, we found that stronger motivational drives accelerated goal-directed behavior: as the motivational need for food increased from 1 to 10 (with water fixed at 1), the time required to reach the food decreased (Fig. 4j).

Offline replay propagates salience information to unexperienced paths

We next investigated how offline learning enhances the network's prediction of salient outcomes and improves behavior. Specifically, we hypothesized that replay activity propagates salience information across the maze to positions, including the ones that were never sequentially visited during online exploration²⁷ (Fig. 5a). To test this, we simulated an offline phase corresponding to when an animal consumes a food reward by providing place input for state 4 to CA3 cells, then observed the resulting spontaneous network activity. During the offline phase, replay activity of place cells spontaneously emerged (Fig. 5b, top). Notably, these replay events included not only previously experienced trajectories, such as left- and right-arm replays, but also a novel "shortcut replay" connecting the two arms that had never been physically traversed in sequence (Fig. 5b, bottom).

We then examined how each replay type contributed to propagating salience information from the reward location to

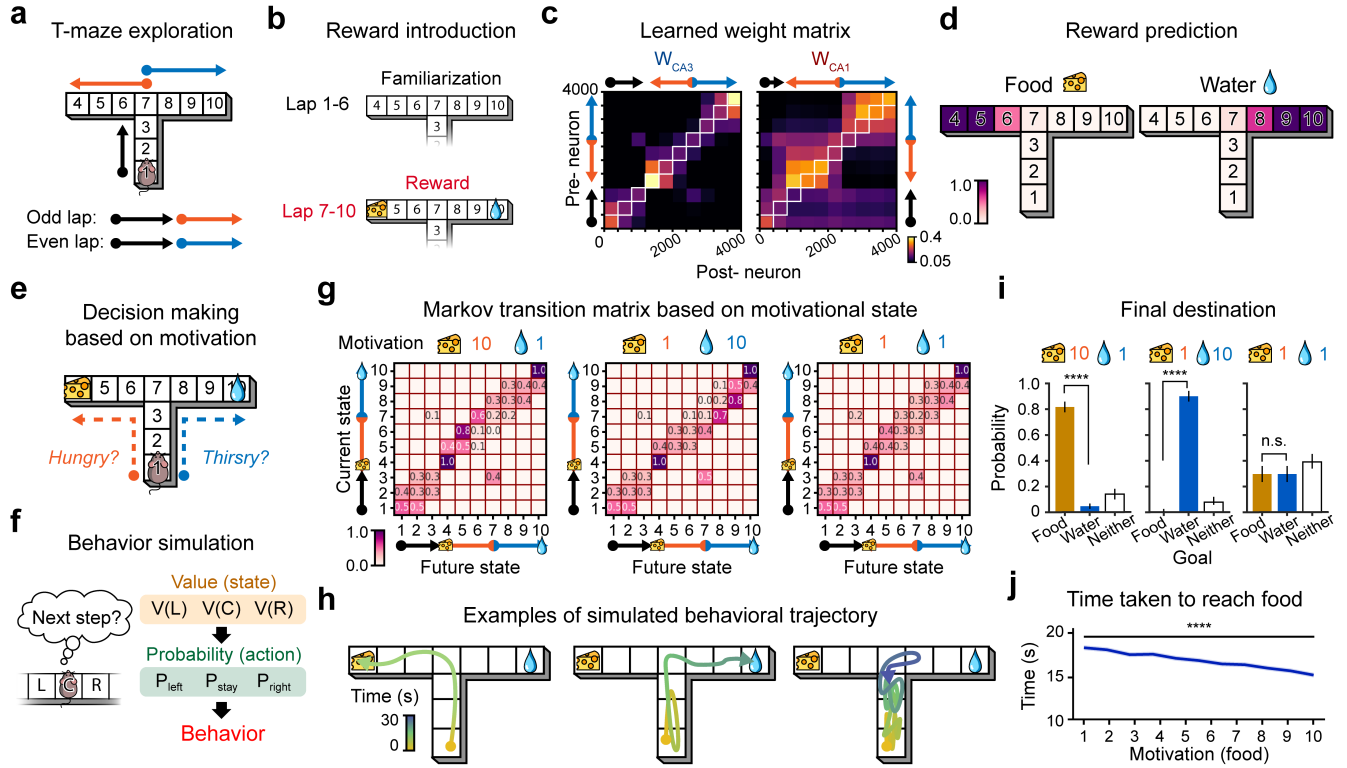


Figure 4. a, T-maze learning scheme. b, After 6 familiarization laps, two distinct rewards (food and water) are introduced at each arm for 4 laps. c, Resulting W_{CA3} and W_{CA1} after lap 10. d, Predictions for food and water across maze positions. e, Animal's decision to pursue either arm of the maze based on its motivational state. f, Behavioral decision-making simulation scheme. g, Markov state transition matrices under three different motivational scenarios for food and water. h, Example simulated trajectories for each scenario. i, Goal reached within 30 seconds of movement for each scenario. Paired t-test, $N_{seed} = 10$, **** $P < 0.0005$, n.s. $P = 0.98$. Data are presented as mean $\pm$ s.d. j, Time to reach the goal as a function of motivational strength. Mann-Kendall test, $N_{trial} = 500$, **** $P = 5.16 \times 10^{-5}$, $Z = -4.05$. Data are presented as mean $\pm$ s.e.m.

distant states. To do this, we updated the network using a symmetric STDP rule³², which adjusts synaptic weights after every spike but with smaller magnitude than BTSP (Fig. 5c). After a single shortcut replay event, food prediction increased across both arms of the maze, with a particularly notable increase at state 10, which had not previously anticipated food (Fig. 5d, left). Left-arm replay enhanced food prediction along its trajectory although the effect was modest because these paths were already experienced during online learning (Fig. 5d, middle). Finally, right-arm replay improved food prediction along its trajectory even though it did not include the food state by leveraging state 7's associations to food (Fig. 5d, right). Together, these findings reveal three principles of replay-mediated prediction improvement: First, predictions increase primarily along the replayed trajectory (Fig. 5e). Second, states that were not previously linked to food show the greatest relative gains. Finally, replay can improve food prediction even when the replayed path does not include the food state, by exploiting connections to already food-associated neighboring states.

We next tested how this replay-induced propagation of salience affected behavior. In the first scenario, the food reward was positioned at the left arm (state 4), and the animal started at the stem (state 1) with a motivational need of 5.0 (Fig. 5f, left). Each trial ended when the animal reached either

end of the maze or failed to reach a goal within 30 steps. We found that including offline learning significantly increased the probability of reaching the food goal (Fig. 5f, right).

Strikingly, offline learning also enhanced adaptability when starting from a novel location. We simulated a paradigm where the animal began at the right end of the maze (state 9), choosing between two possible goals (state 4 and state 1) with the food still located at state 4 (Fig. 5g, left). The inclusion of an offline learning phase again increased the likelihood of reaching the reward within the time limit (Fig. 5g, right). We further demonstrated that the stronger spatial input during the offline phase increases the frequency of replay events (Supplementary Fig.??). The intensity of the spatial input during the offline phase can possibly be modulated by perceived salience, as proposed by previous studies³⁶.

The model learns aversive outcomes, predicts their presence through forward rollouts, and exhibits avoidance behavior

We next asked whether the model could also account for the learning of aversive features. This differs from reward learning (Fig. 3–5) because animals spend minimal time at aversive sites, limiting the opportunity to develop over-representation through prolonged dwell time. Moreover, "on-site" replay, where salience information is propagated during offline learn-

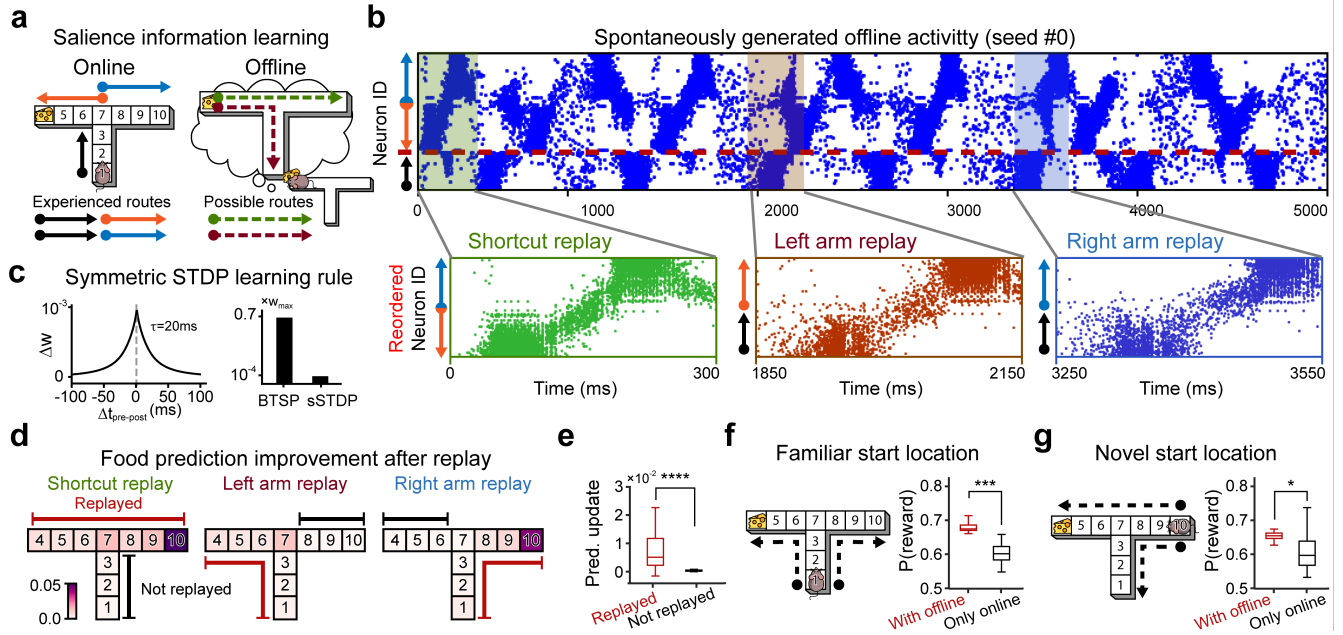


Figure 5. **a**, Propagation of salience information through online and offline learning. **b**, Spontaneously generated offline activity. Replay events are grouped into three categories based on their trajectories: shortcut, left-arm, and right-arm. **c**, Left: sSTDP rule used in offline learning. Right: Largest possible weight update of BTSP and sSTDP rules. **d**, Normalized food prediction update in state-wise reward prediction. The normalized update was computed by first subtracting the pre-update prediction from the post-update prediction, and then dividing this difference by the pre-update prediction. **e**, Normalized food prediction update of states that were replayed (red) and the those that were not replayed (black). Independent t-test, $N_{\text{replayed}} = 70$ vs. $N_{\text{notreplayed}} = 30$, $****P = 2.28 \times 10^{-4}$. **f**, Behavioral simulation results when the animal starts from the same location as during the online phase. Paired t-test, $N_{\text{seed}} = 10$, $***P = 3.72 \times 10^{-3}$. **g**, Behavioral simulation results when the animal starts from a different location than in the online phase. Paired t-test, $N_{\text{seed}} = 10$, $*P = 0.01$. All data are presented as mean $\pm$ s.d.

ing while the animal remains at the location, is not possible. At the behavioral level, even a single exposure to a negative outcome typically repels the animal, reducing future visits and further constraining learning.

To simulate this, we followed the design of a previous study⁹, in which mice freely explored a linear track before encountering a shock at the track end. In our simulation, the animal completed three laps without salient outcomes (from state 1 to 8 and back), followed by a fourth lap in which a shock occurred at state 8 (Fig. 6a). During the shock, the animal's speed doubled and the motivational importance of the shock was set to 10 (Fig. 6b). Compared to familiarization laps, perceived salience sharply increased at state 8 during shock exposure (Fig. 6c), inducing over-representation of that location despite the short dwell time (Fig. 6d).

We then simulated an offline phase in which the animal was removed from the environment by providing random, low-intensity input to CA3 cells for 10 seconds (Fig. 6e). As a result, spontaneous replay-like activity emerged, spanning the entire track (Fig. 6f), upon which the network was updated using the symmetric STDP rule³².

We found that incorporating offline learning extended the "shock prediction field": replay enabled the shock to be predicted from locations further away along the track (Fig. 6g). This change in prediction translated into altered behavior. With an offline learning, the animal began to avoid moving toward the shock zone from more distant states, including

states 2-4 (Fig. 6h). Simulating 600 s of free movement revealed that offline learning reduced the animal's tendency not only to enter the shock zone (state 8), but also to occupy near-shock states (states 5-7) (Fig. 6i). We quantified this effect using state occupancy: incorporating an offline learning phase decreased both shock-zone occupancy (Fig. 6j, left) and near-shock occupancy (Fig. 6j, right). This is consistent with the study we reference, in which shock exposure reduced the animal's tendency to visit the half of the track near the shock location⁹.

An important finding from this study was the observation that, when the mice paused as they walked toward the shock zone after reintroduction to the environment, they exhibited forward-directed replay events toward the shock zone⁹. We hypothesized that such "forward roll-outs" allow the animal to mentally access information about the shock without physically entering it. To test this, we simulated an offline phase with place input from the middle of the track (state 4) (Fig. 6k). The network spontaneously generated sequential place cell activity extending toward both ends of the track (Fig. 6l, left), including forward roll-out events that included the shock state (Fig. 6l, right). Consistent with our hypothesis, internal shock prediction increased during these roll-out events (Fig. 6m).

Together, these results show that aversive stimuli can be learned reliably despite minimal exposure and suggest potential roles for both off-site and on-site replay in supporting avoidance behavior.

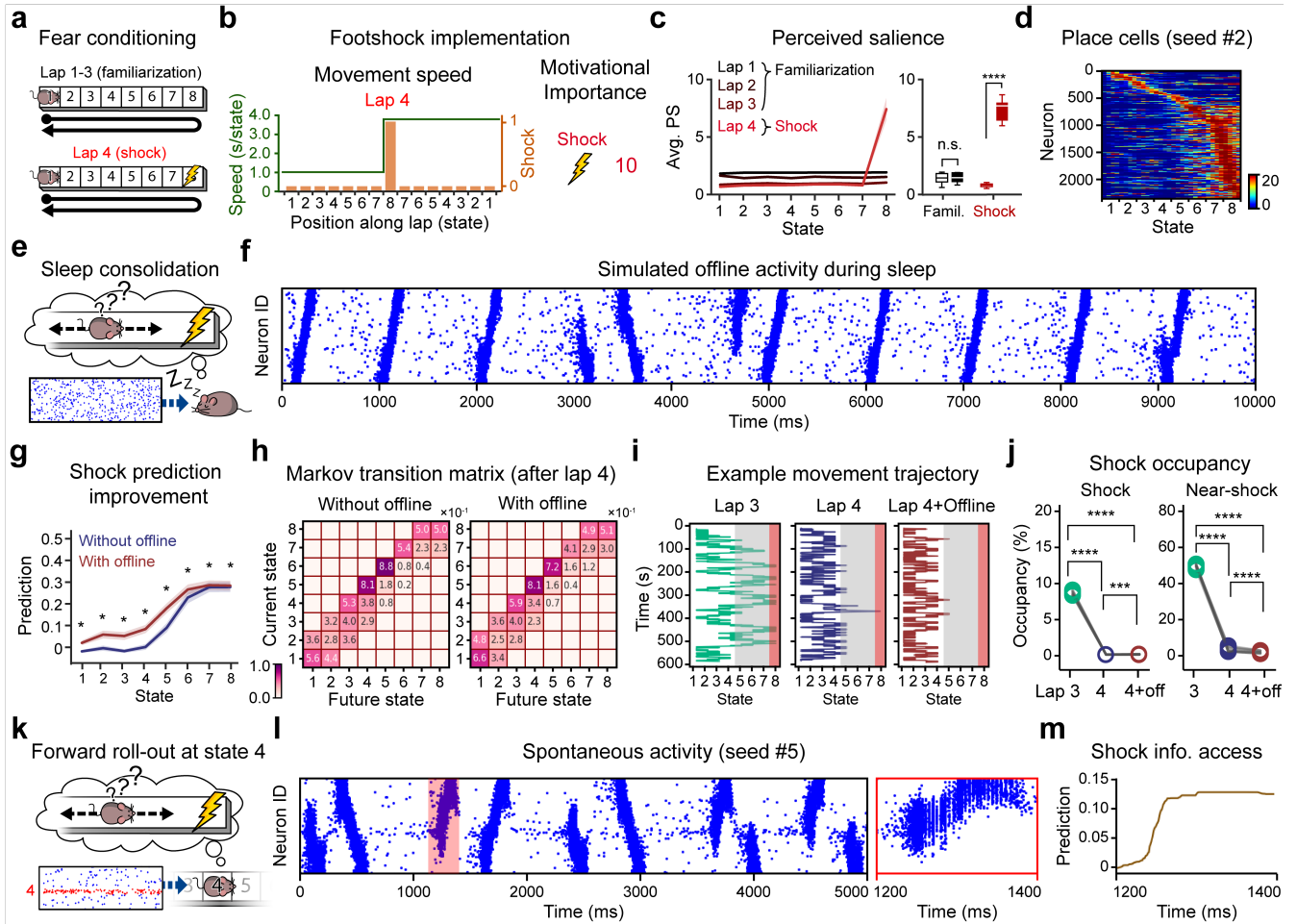


Figure 6. a. Aversive Wafeature learning scheme. **b.** Two factors modulated by the introduction of aversive outcome: increased movement speed and the motivational importance of the aversive outcome. **c.** Average perceived salience. Perceived salience of each state on lap 1 to 4. The states without salient outcome does not show significant difference across different laps, whereas the perceived salience of the punished state significantly increased after the introduction of the footshock. Independent t-test, perceived salience of the non-salient states vs. perceived salience of the salient state, $N_{seed} = 10$, n.s. $P = 0.49$, $****P = 3.41 \times 10^{-64}$. **d.** CA1 place cells clustered around the punished state, over-representing it. **e.** Offline simulation at state 4. **f.** Example spontaneously generated offline activity (10 s). **g.** Elevated anticipated shock stimulus after the offline learning phase. Paired t-test, shock prediction at each state without offline learning vs. with offline learning, $N_{seed} = 10$, $*P < 3.00 \times 10^{-7}$. **h.** Markov state transition matrices before and after the offline learning. **i.** Simulated movement trajectories before shock introduction (left), after shock introduction without offline learning **j.** Left: Shock state occupancy of each scenario. Paired t-test, $N_{seed} = 10$, $***P = 4.84 \times 10^{-4}$, $****P < 6.82 \times 10^{-15}$. Left: Occupancy of the right half of the track, including the shock state. Paired t-test, $N_{seed} = 10$, $****P < 5.71 \times 10^{-5}$. **k.** Forward roll-out at state 4, simulated by inputting state 4-specific input. **l.** Left: Example spontaneously generated offline activity (5 s). Right: Zoomed-in roll-out activity. **m.** Shock presence prediction output from feature-prediction layer during the roll-out activity. All data are presented as mean $\pm$ s.d.

Discussion

In this study, we introduced a unified normative theory of the cognitive map and its role in value-based decision making, bridging from optimality considerations to representations, learning rules and behavior. Our framework extends the successor representation by incorporating successor features, and draws BTSP and replay to address the rapid adaptations seen in hippocampal circuits and behavior. We showed how this framework can be implemented in the hippocampus: CA3 encodes a salience-neutral representation of environmental structure, CA1 integrates this with salience information to form a PS-weighted successor representation, and a downstream feature-prediction layer computes successor features. These predicted features can then be combined with context-varying values to guide decisions. The model is learned through BTSP during online exploration and further refined by symmetric STDP during offline replay. It adapts swiftly when environmental contingencies change and captures a range of experimentally observed behaviors, linking synaptic mechanisms to network-level representations and flexible actions.

Our framework advances normative accounts of the cognitive map, particularly SR, by explicitly separating salience from value. This distinction allows the model to explain why hippocampal networks over-represent salient locations, even when those locations are not inherently rewarding and do not lead to increased future visitation (Fig. 3 and 6). In addition, incorporating SF enables flexible integration of time-varying information such as motivational needs¹⁰. Because the model predicts features rather than states *per se*, it can rapidly adjust subjective value related to the features and adapt behavior based on motivational state, without requiring relearning of the underlying environmental structure (Fig. 4).

Mechanistically, our work incorporates and extends prior findings on BTSP's role in driving over-representation of rewarding locations^{18,21}. We further model how this over-representation may arise through different factors for rewarding versus aversive stimuli. In our simulations, positive outcomes were learned through moderate salience signals combined with prolonged occupancy (Fig. 3), whereas aversive outcomes were encoded through much stronger salience despite extremely brief dwell times (Fig. 6). Our model also predicts how these over-representations accelerate learning of feature–location associations, as elevated activity at salient sites drives faster acquisition within the feature-prediction layer (Fig. 3).

Our model proposes a unified account of replay's contribution to learning the cognitive map by linking existing findings. Recent work suggests that CA3 learns through BTSP with a symmetric time kernel³¹, that symmetric recurrent weight profiles can generate spontaneous replay^{32,37}, and that replay contributes to consolidating SR-like representations³⁸. Integrating these findings, our model shows that a symmetric, value-neutral map of environmental structure emerges in CA3 after a single lap of exploration via symmetric BTSP, which generates spontaneous replay even in the absence of

informative input (Fig. 2, supplementary Fig. ??). These replay events, in turn, strengthened the cognitive map to facilitate earlier prediction of salient features (Fig. 5 and 6). Moreover, because CA3 encodes structural relationships independently of actual experience, the network can generate replay sequences along physically unexplored trajectories^{27,29} (Fig. 5). This enables replay-driven inference: the salience information is propagated to states never sequentially paired with salient features and improves predictions at otherwise unrelated locations. Finally, we show that the strength of upstream input to CA3 during the offline phase, potentially modulated by the salience of the experienced state¹⁹, controls the frequency and length of replay events (Supplementary Fig. ??). This is consistent with experimental reports that salience boosts replay expression³⁶.

Our model also accounts for a role of replay in value computation that leads to adaptive behavior. Simulating the offline phase within the track, analogous to immobile period during online exploration, produced forward roll-outs similar to those observed experimentally⁹ (Fig. 6). Our model predicts that these roll-outs transiently activate representations of non-current states, particularly salient ones, enabling the prediction of features and the estimation of state values required for planning^{33,39,40}.

Nevertheless, important questions remain. For example, the relative importance of replay in strengthening newly acquired salient memories with maintaining previously learned information to prevent catastrophic interference during new learning^{41,42} remain ongoing topics for discussion and debate. In addition, we did not explore how replay content is selected or modulated^{10,43}. Future work should examine these issues in greater detail to clarify how replay contributes to learning, memory, and adaptive behavior.

Lastly, our framework also has implications for artificial intelligence (AI). Current AI systems often require large datasets and struggle with generalization, in part because their learning procedures make it difficult to extract and reuse structural regularities across tasks. In contrast, our approach suggests that rapid online learning through BTSP and offline refinement through replay may offer alternatives to standard gradient-based updates and help address these limitations^{44,45}. Specifically, our framework that separates learning about environmental structure from learning about features and assigning value to them³⁰, enabling rapid adaptation to changes in environmental contingencies and flexibility under shifts in internal motivational state. Replay further expands the scope of learning by propagating information to unvisited states, enhancing generalization without additional experience. In our T-maze simulations, for example, the network reproduced a key experimental finding^{27,29} by inferring connections between unexplored paths and recovering the maze's structural layout. Finally, replay in our model emerges spontaneously, without external supervision, continually strengthening previously-acquired internal representations. This suggests a path toward AI systems that autonomously improve their internal repre-

sentations during "rest" periods, thereby achieving greater efficiency and adaptability.

Taken together, we propose an overarching framework for the cognitive map in which BTSP-based online learning and offline replay jointly support adaptive prediction and decision making. We showed how this framework applies to different behavioral tasks, including T-maze navigation and shock avoidance. By connecting synaptic learning rules to network-level representations and behavior, our work provides a plausible, efficient, and adaptive account of how animals learn, generalize, and make decisions in dynamic environments.

Methods

Neural network model

The hippocampal network

We modeled the hippocampal network as a two-layer spiking neural network, corresponding to the CA3 and CA1 subregions of the hippocampus. Consistent with experimental findings, the CA3 layer included recurrent connectivity (W_{CA3}) and projected feedforward output to CA1 (W_{CA1}). Synaptic weights in both pathways were constrained between 0 and 10, initialized from a Gaussian distribution (mean = 0.01, standard deviation = 10^{-4}), and formed with a fixed connection probability of 0.1.

Both CA3 and CA1 layers contained the same number of excitatory neurons, determined by the number of states in the simulated maze. Each state was represented by 400 neurons, resulting in 3,200 excitatory neurons per layer for the linear track (8 states) and 4,000 neurons per layer for the T-maze (10 states). CA3 excitatory neurons were designated as place cells, with uniformly distributed place fields and spatial inputs reflecting the animal's position during online exploration. CA1 excitatory neurons did not receive direct spatial input; instead, their place-selective responses emerged through learned W_{CA1} .

Inhibitory modulation differed between online and offline phases. During online exploration, a uniform feedback inhibition signal was applied to all excitatory neurons within each layer to modulate the probability of plateau potential (PP) events. During the offline replay phase, 150 inhibitory interneurons were explicitly included in each layer to support spontaneous replay dynamics.

Feature prediction network

The feature prediction network aimed to predict the physical presence of environmental features, including background, food, water, and shock. It comprised units representing the predicted presence (ψ) of each feature. These units received fully connected input from CA1 neurons via W_{pred} and sent a novelty signal (N) back to CA1, modulating the likelihood of plateau potentials in CA1 cells. We interpret this network as a simplified representation of brain circuits involved in feature prediction and value computation, such as the nucleus

accumbens (NAc), which forms a recurrent loop with CA1 and relays dopaminergic signals.

Online exploration

We simulated rodent's maze learning task which is commonly studied in experimental literature. On each lap, the animal followed a sequence of movements, and the hippocampal and feature prediction networks were trained accordingly.

Maze exploration scenarios

We simulated three different maze learning scenarios:

- **An animal unidirectionally navigating a linear-track treadmill¹⁸:** Each complete rotation of the belt corresponded to a 2.4 m lap. For convenience, the track was discretized into 8 states (state 1 to 8, from left to right), each 30 cm in length. In each lap, the animal began at the leftmost state (state 1), traversed to the rightmost state (state 8), reached the end of the belt, and was returned to the starting point. Formally,

$$\mathcal{S}_{\text{lap}} = \langle 1, 2, \dots, 8 \rangle.$$

The next lap began immediately after the animal returned to state 1.

- **An animal navigating a T-maze¹⁰:** The maze consisted of a 0.9 m vertical stem (state 1 to 3, from bottom to top), connected at its midpoint to a 2.1 m horizontal arm (state 4 to 10, from left to right, with state 7 as the intersection). In each lap, the animal began at the base of the stem (state 1), moved up to the center of the horizontal arm (state 7), and then turned left in odd-numbered laps (ending at state 4) or right in even-numbered laps (ending at state 10).

Formally,

$$\mathcal{S}_{\text{lap}} = \begin{cases} \langle 1, 2, 3, 7, 6, 5, 4 \rangle, & \text{if } N \text{ is odd,} \\ \langle 1, 2, 3, 7, 8, 9, 10 \rangle, & \text{if } N \text{ is even.} \end{cases} \quad (7)$$

- **An animal round-tripping a linear maze⁹:** This scenario replicates an animal exploring a confined, 2.4 m segment of linear maze. The track was discretized into 8 states (state 1 to 8, from left to right), but unlike the linear treadmill, state 8 and state 1 are not connected. A single lap began with the animal at the left end (state 1), moving sequentially to the right end (state 8), and then returning back to the start (state 1). Formally,

$$\mathcal{S}_{\text{lap}} = \langle 1, 2, \dots, 7, 8, 7, \dots, 1 \rangle.$$

Environmental features

In our simulation, states were associated with distinct environmental features, f . Environmental features are accompanied with their physical intensity I_f as well as motivational importance MI_f . The presence and absence of the features at a certain state s were encoded as a presence vector $P(s)$. Although

real maze exploration environments contain many features, we simplified this to capture essential elements relevant to our work: background, food, water and shock.

- **Background:** Each state involves state-specific background feature, which we thought of as sensory cues that enables the animal to distinguish the state from other ones. Since this feature is not attached to any affective value, we set the MI of this feature to be 1. When the state only has background but not other significant environmental features, the animal moves at the baseline speed, taking $T = 1$ second.
- **Food and water:** These features serves as a positive outcome in our simulation, and has MI of 3. When the animal was passing the state with these features, it slowed to half the baseline speed, simulating an extended stay during reward consumption ($T = 2$ s).
- **Shock:** This feature serves as a negative outcome, with MI of 10. When the animal passes a state involved with shock, it sped up to twice the baseline speed, reflecting an aversive response ($T = 0.5$ s).

In our simulation, we didn't vary the physical intensity much. Simply, I_f was set to be 2 if f is present, and 1 if it is absent. i.e., in our simulation, $I_f = P_f + 1$.

Another component attached to environmental features is novelty, denoted as $N_f(t)$. This quantity tracks the animal's subjective surprise associated with feature f , and defined as the difference between the predicted presence and actual presence of the feature. The novelty values were updated at each timestep based on the absolute prediction error computed by the feature prediction network, ϵ , as described in the following subsection.

$$N_f(t) = |\epsilon_f(t)| \quad (8)$$

Computation of perceived salience

The perceived salience (PS) of an environmental feature f , denoted as PS_f , represents the animal's subjective evaluation of how noticeable or memorable f is. In our simulations, PS was modulated by three factors: motivational importance MI , intensity I , and the novelty N of the environmental feature. Specifically, the perceived salience of feature f , PS_f , was then computed as:

$$PS_f(t) = I_f \cdot MI_f \cdot N_f(t) \quad (9)$$

The perceived salience of a state at time t , $PS(t)$, was computed as the sum of PS across all environmental features.

Spike train generation and activity level estimation

CA3 place fields were uniformly distributed along the x - and y -axes, with each field centered at a distinct position. Place fields were circular with a radius of 30 cm.

Spiking activity in CA3 excitatory neurons arose from two sources: external spatial input from the entorhinal cortex and recurrent input from other CA3 neurons. Firing driven by spatial input (FR_{place}) depended on the animal's Euclidean distance from the center of a neuron's place field and followed a Gaussian profile. The maximum value of FR_{place} was set to 20 Hz and the standard deviation of the Gaussian was chosen to be the radius of the field.

Upstream input contributed an additional firing component ($FR_{\text{recurrent}}$), calculated as the sum of activities from all connected presynaptic neurons, weighted by W_{CA3} . This input was then passed through a sigmoid nonlinearity to yield the final recurrent input-driven firing rate for each neuron. The sigmoid function was defined as:

$$\sigma(x, a, b) = \frac{1}{1 + \exp(-b(x - a))}, \quad (10)$$

where $a = 4$ and $b = 3$.

Finally, the total firing rate of a CA3 neuron i was given by:

$$FR_i = FR_{\text{place},i} + FR_{\text{recurrent},i}. \quad (11)$$

CA1 excitatory neurons were modeled similarly but did not receive direct spatial input. Their firing rates were determined exclusively by feedforward input from CA3: the activity of CA3 presynaptic neurons was multiplied by W_{CA1} and passed through the same sigmoid function with $a = 4$ and $b = 2$ to compute $FR_{\text{feedforward}}$. Therefore, the total firing rate of a CA1 neuron j is:

$$FR_j = FR_{\text{feedforward},j}. \quad (12)$$

After calculating FR s of all neurons in both CA3 and CA1, we generated spike trains for each neuron using inhomogeneous poisson processes. These spike trains were then filtered to remove spikes that violated the 5 ms refractory period.

Spike trains for both CA3 and CA1 were generated at every Δt_{spike} interval, which was set to 100 ms. For CA3 spike train generation, the input activity level was determined from the CA3 spike train generated in the previous time block.

To quantify neuronal activity, we calculated the actual average firing rate of each neuron by dividing the total number of generated spikes by Δt_{spike} . Note that this differs from the ideal FR , as it reflects post-processing by both the inhomogeneous poisson process^{32,37} and the refractory period filter.

Training the feature prediction network using the delta rule

The feature prediction network was trained during the on-line exploration phase using the error-correcting delta rule³⁵. Specifically, a synaptic weight w_{fi} connecting a CA1 neuron i and a feature unit f , was updated as:

$$\Delta w_{fi}(t) = \mathcal{A}_i(t) \cdot \epsilon_f(t) \quad (13)$$

where $\mathcal{A}_i(t)$ is the activity of neuron i and $\epsilon_f(t)$ is the prediction error of feature f at time t . The prediction error was

defined as the error between predicted presence (ψ) and the physical presence (ϕ) for that feature:

$$\varepsilon_f(t) = \psi_f(t) - \phi_f, \text{ where} \quad (14)$$

$$\psi_f(t) = \sum_i \mathcal{A}_i(t) \cdot w_{ji}(t) \quad (15)$$

Note that the absolute value of the prediction error of a feature is the novelty N associated with the feature.

Training the hippocampal network using BTSP

Synaptic weight updates in the hippocampal network followed a BTSP rule, which depends on the interaction between presynaptic eligibility traces (ETs) and postsynaptic instructive signals (ISs). ETs were computed based on the spiking activity of presynaptic neurons, while ISs were driven by the postsynaptic neuron's dendritic plateau potentials (PPs).

ETs were calculated only for CA3 neurons, as these served as the sole presynaptic population in our model. When a presynaptic spike occurred, the neuron's ET increased by a fixed amplitude, A_{ET} , and then decayed exponentially with a time constant τ_{ET} .

ISs were evoked by plateau potentials (PPs), modeled as stochastic events occurring in both CA3 and CA1 neurons. Each neuron k was assigned a PP probability $p_{\text{plateau},k}$, initialized to a basal value p_{basal} and updated every Δt_{spike} . In our simulations, p_{basal} was set to 0.05. At each 1 ms time step, a Bernoulli trial was performed using $p_{\text{plateau},k}$ to determine whether neuron k initiated a PP. The probability of PP occurrence, $p_{\text{plateau},k}$, was enhanced by the neuron's recent activity and the perceived salience (PS) of the current state, but decreased with normalized population activity $\mathcal{A}_{\text{norm}}$, defined as the mean activity across neurons within the layer^{21,22} (Supplementary Fig.??h). Formally,

$$p_{\text{plateau},k} = \mathcal{A}_k \cdot c_{\text{norm}}, \quad (16)$$

where

$$c_{\text{norm}} = \begin{cases} p_{\min} + 2p_{\text{basal}} \cdot \sigma(\mathcal{A}_{\text{target}} - \mathcal{A}_{\text{norm}}, 0, 5), & \text{if } \mathcal{A}_{\text{target}} > \mathcal{A}_{\text{norm}}, \\ p_{\min}, & \text{otherwise.} \end{cases} \quad (17)$$

When a PP occurred, the postsynaptic neuron's instructive signal increased by A_{IS} and decayed exponentially with a time constant τ_{IS} . BTSP-mediated synaptic updates occurred only during PP events. The BTSP update of the weights is a function of $ET \times IS$ as well as the initial value of the weight²¹. Initially weak weights undergo stronger potentiation even with smaller $ET \times IS$, whereas the weights that are already strong enough it more likely to undergo depression even with stronger ET and IS signal. Formally,

$$\Delta w_{ji} = (w_{\max} - w_{ji}) k_{\text{pos}} c_{\text{pos}} - w_{ji} k_{\text{neg}} c_{\text{neg}}, \quad (18)$$

where

$$c_{\text{pos}} = \frac{\sigma(ET_i \cdot IS_j; a_{\text{pos}}, b_{\text{pos}}) - \sigma(0; a_{\text{pos}}, b_{\text{pos}})}{\sigma(1; a_{\text{pos}}, b_{\text{pos}}) - \sigma(0; a_{\text{pos}}, b_{\text{pos}})}, \quad (19)$$

$$c_{\text{neg}} = \frac{\sigma(ET_i \cdot IS_j; a_{\text{neg}}, b_{\text{neg}}) - \sigma(0; a_{\text{neg}}, b_{\text{neg}})}{\sigma(1; a_{\text{neg}}, b_{\text{neg}}) - \sigma(0; a_{\text{neg}}, b_{\text{neg}})}. \quad (20)$$

In our simulations, we set $k_{\text{pos}} = 0.085$, $k_{\text{neg}} = 0.01$, $a_{\text{pos}} = 0.8$, $b_{\text{pos}} = 6$, $a_{\text{neg}} = 0.05$, and $b_{\text{neg}} = 44.44$.

Offline phase

The offline simulation framework was adapted from Ecker et al.³² Here, neurons were modeled using adaptive exponential integrate-and-fire dynamics. Hippocampal weights learned during online exploration (W_{CA3} and W_{CA1}) were carried over into the offline phase, with additional connectivity to inhibitory interneurons introduced. Spontaneous activity emerging in the CA3 and CA1 layers during this phase was then used to update W_{CA3} and W_{CA1} according to a symmetric STDP rule.

Connectivity optimization

Additional connectivity was introduced and optimized using an evolutionary algorithm, including: entorhinal cortex input to CA3 excitatory cells ($\text{EC} \rightarrow \text{Ext}_{\text{CA3}}$); CA3 excitatory cell to CA3 inhibitory interneuron ($\text{Ext}_{\text{CA3}} \rightarrow \text{Inh}_{\text{CA3}}$); interneuron–interneuron ($\text{Inh}_{\text{CA3}} \rightarrow \text{Inh}_{\text{CA3}}$); and inhibitory interneuron to excitatory cell ($\text{Inh}_{\text{CA3}} \rightarrow \text{Ext}_{\text{CA3}}$). The same connectivity types were established in CA1 ($\text{Ext}_{\text{CA1}} \rightarrow \text{Inh}_{\text{CA1}}$, $\text{Inh}_{\text{CA1}} \rightarrow \text{Inh}_{\text{CA1}}$, $\text{Inh}_{\text{CA1}} \rightarrow \text{Ext}_{\text{CA1}}$).

The optimization algorithm minimized a loss function designed to match network-level physiological targets: maintaining realistic population firing rates in excitatory cells, suppressing gamma-band (30–100 Hz) oscillations, and enhancing ripple-band oscillations in excitatory cell populations.

During the offline phase, spontaneous activity was simulated in both CA3 and CA1, with low-rate entorhinal cortex input to CA3. W_{CA3} and W_{CA1} were updated using a symmetric STDP rule throughout this period.

The detailed description of spiking neural network model along with its parameters are described in supplementary materials.

Training the hippocampal network using symmetric STDP

Symmetric STDP was used to update network using the spike train generated during the offline phase. It only involves potentiating the connections which connect neurons that fire together. Precisely, the update rule we used is described as follows, with $A_{\text{STDP}} = 1 \times 10^{-3}$ and $\tau_{\text{STDP}} = 20\text{ms}$.

$$\Delta w = A_{\text{STDP}} \cdot e^{|t_{\text{post}} - t_{\text{pre}}| / \tau_{\text{STDP}}} \quad (21)$$

Behavioral simulation

After the network learned the environment including its spatial structure and associated features, we evaluated its performance

in behavioral decision-making tasks. This phase corresponds to the exploitation stage in reinforcement learning, during which the agent uses previously acquired knowledge to make optimal choices.

Decision-making scenarios

We simulated two behavioral decision-making scenarios: one involving a positive outcome in a T-maze and another involving an aversive outcome in a linear track.

- **T-maze with two goal locations and distinct rewards¹⁰**: Each trial began with the animal positioned at the stem of the maze (state 1) and ended when it reached either goal state (state 4 or 10). Food was located at the left end (state 4) and water at the right end (state 10), and the animal's motivational needs for these two rewards were systematically varied.

To test the network's adaptability to different starting conditions, additional simulations were conducted in which the animal started from the right end (state 10) and traversed either toward the left end (state 4) or the stem (state 1), with food placed at state 4.

In all cases, the animal was allowed a maximum of 30 steps per trial. Trials were terminated if the animal failed to reach a goal within this limit, and such cases were labeled as "failure to find the goal."

- **Linear track with an aversive outcome⁹**: Each trial began with the animal positioned at the left end of the track (state 1) and allowed to move freely for 600 steps. A shock stimulus was positioned at the right end of the track (state 8).

Value computation and its usage in decision-making

At each time step, the animal evaluated the value of neighboring states and selected its next action according to a softmax policy. The value of a given state s , denoted $V(s)$, was computed as:

$$V(s) = \sum_f \psi_f(s) \cdot v_f, \quad (22)$$

where $\psi_f(s)$ represents the predicted presence of feature f at state s , and v_f denotes the animal's subjective value of that feature. The feature values v_f were dynamically modulated by the animal's transient motivational state. Given these computed state values, transition probabilities were derived using a softmax function, and the environment was treated as a Markov chain with known state transitions.

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